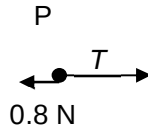
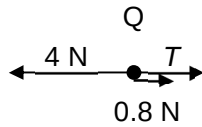


5



$$\begin{aligned} F_{\text{on } Q} &= m_Q a = 4a \\ &= 4 - 0.8 - T \end{aligned}$$

$$\begin{aligned} F_{\text{on } P} &= m_P a = 2a \\ &= T - 0.8 \end{aligned}$$

solve simultaneous:

$$4a = 4 - 0.8 - T$$

$$2a = T - 0.8$$

$$a = 0.4 \text{ ms}^{-2}$$

motion of P relative to Q:

$$a_{\text{relative}} = 2a$$

$$s_{\text{relative}} = 0.40 - 0.15 = 0.25 \text{ m}$$

$$s = ut + \frac{1}{2}at^2 = 0 + \frac{1}{2}at^2$$

$$\begin{aligned} t &= \sqrt{\frac{2s}{a}} = \sqrt{\frac{2(0.25)}{0.8}} \\ &= 0.79 \text{ s} \end{aligned}$$

6